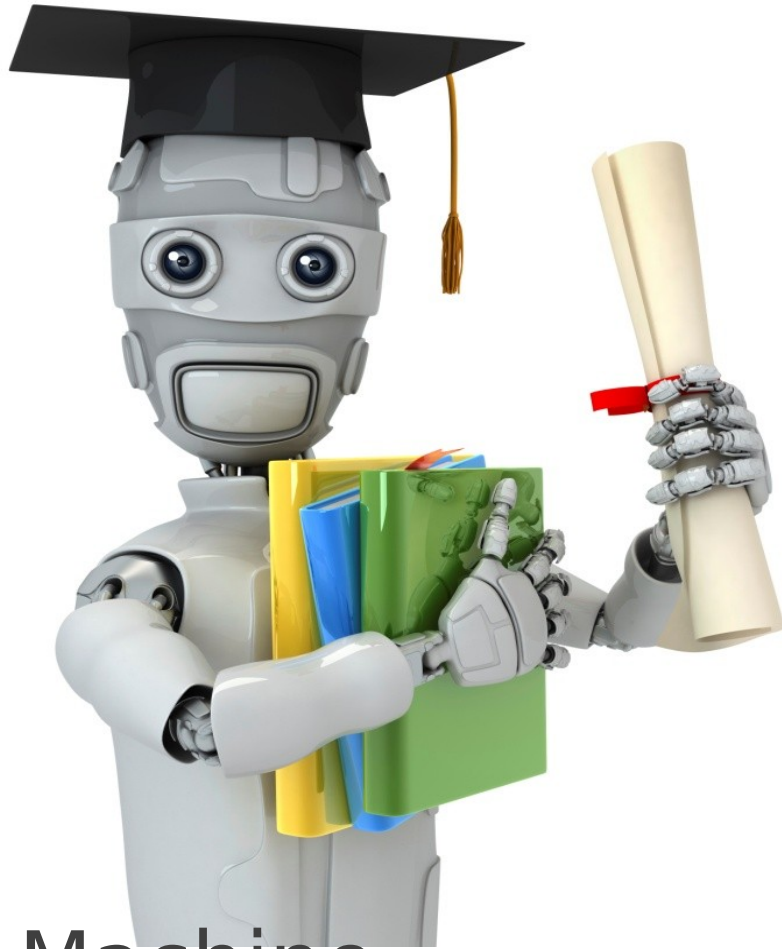




Machine
Learning

Introduction

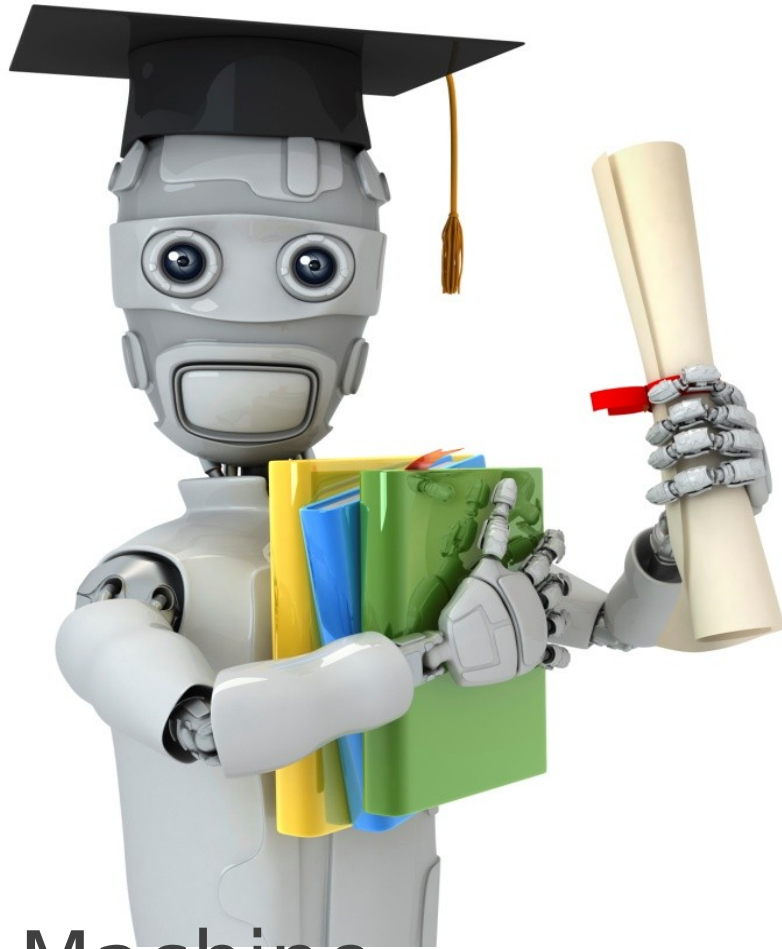
Welcome

Machine Learning

- Grew out of work in AI
- New capability for computers

Examples:

- Database mining
 - Large datasets from growth of automation/web.
 - E.g., Web click data, medical records, biology, engineering
- Applications can't program by hand.
 - E.g., Autonomous helicopter, handwriting recognition, most of Natural Language Processing (NLP), Computer Vision.
- Self-customizing programs



Machine
Learning

Introduction

What is machine learning

Machine Learning definition

- Arthur Samuel (1959). Machine Learning: Field of study that gives computers the ability to learn without being explicitly programmed.
- Tom Mitchell (1998) Well-posed Learning Problem: A computer program is said to *learn* from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .

“A computer program is said to *learn* from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .”

Suppose your email program watches which emails you do or do not mark as spam, and based on that learns how to better filter spam. What is the task T in this setting?

Classifying emails as spam or not spam.

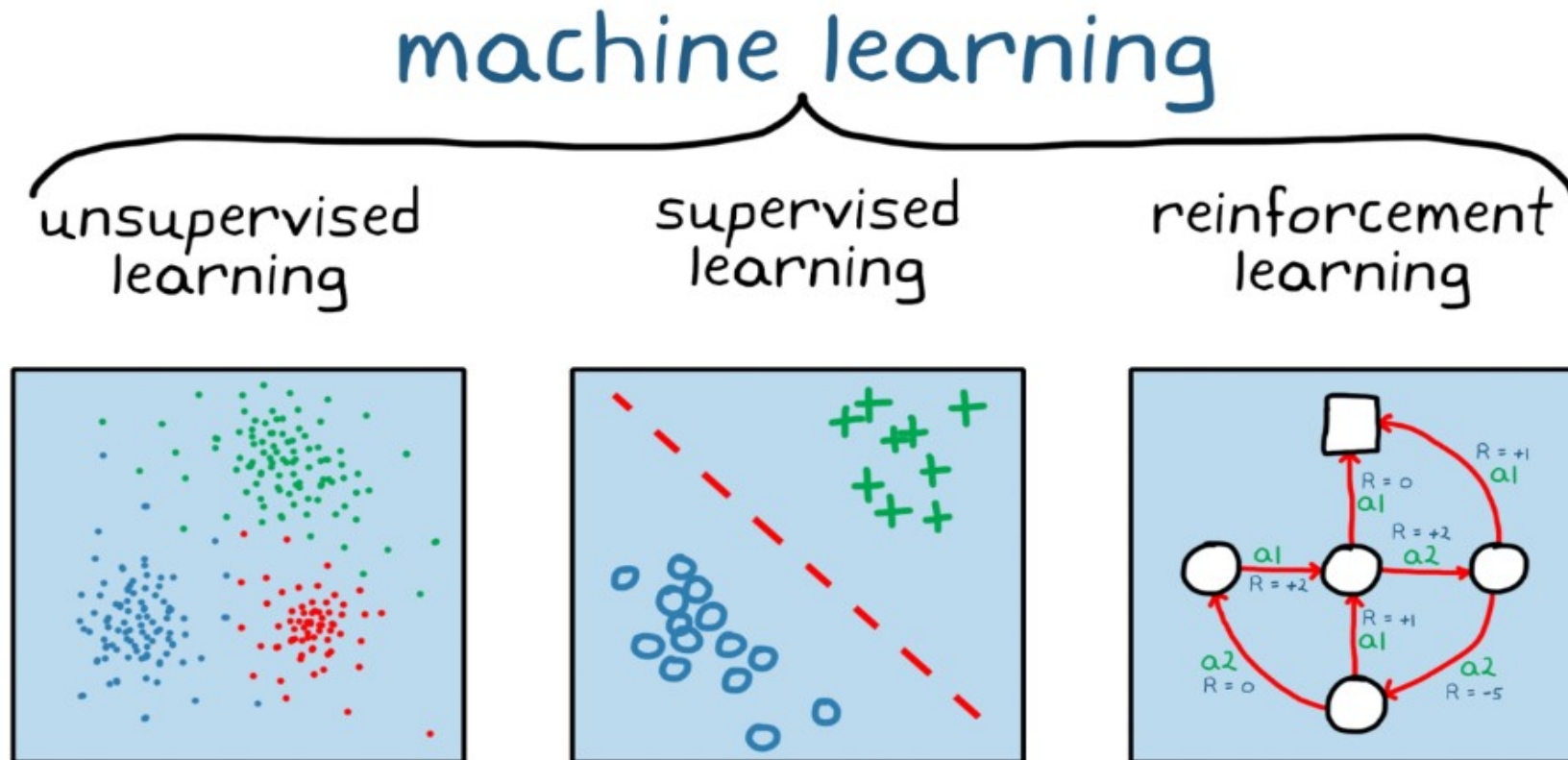
Watching you label emails as spam or not spam.

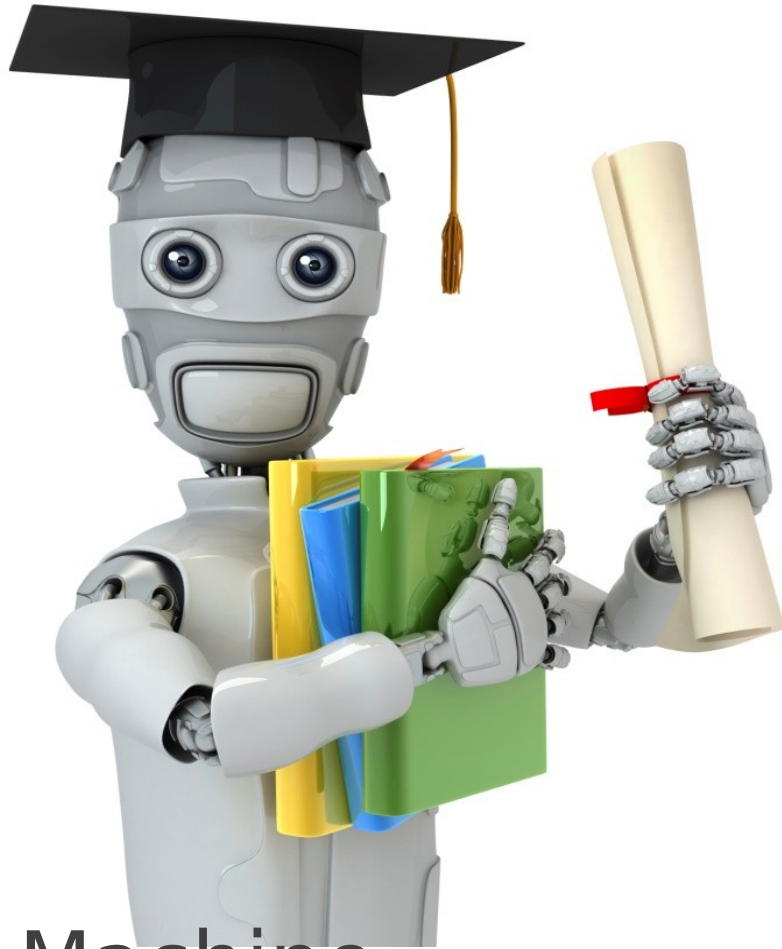
The number (or fraction) of emails correctly classified as spam/not spam.

None of the above—this is not a machine learning problem.

Machine learning algorithms:

- Supervised learning
- Unsupervised learning
- Reinforcement learning





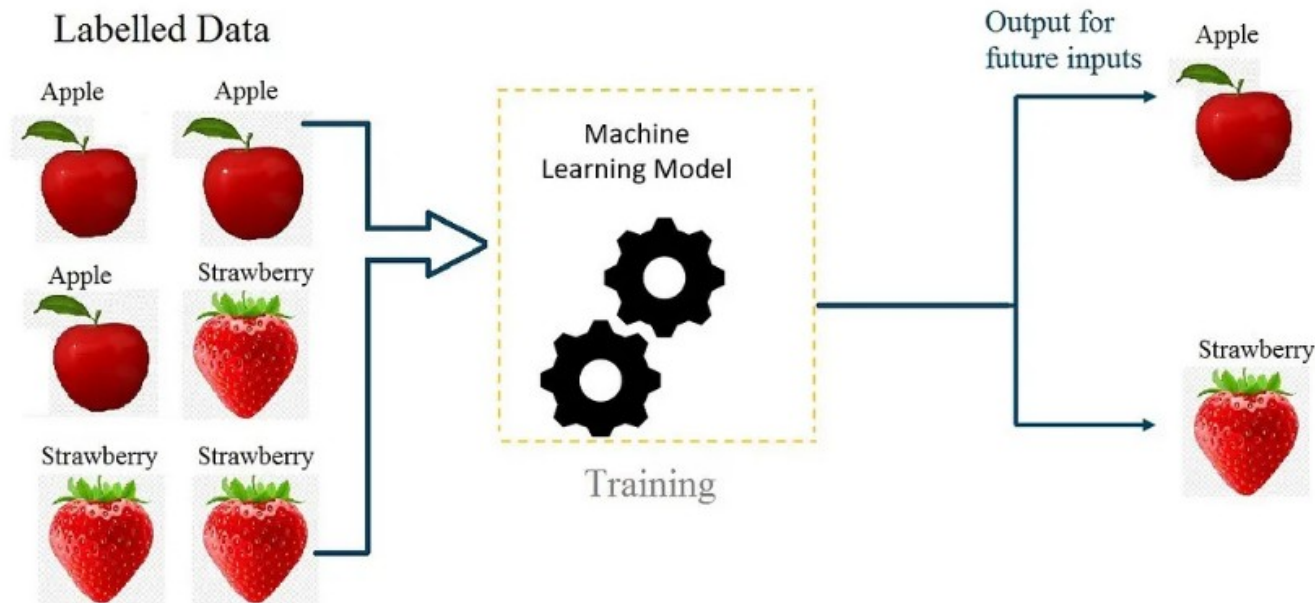
Machine
Learning

Introduction

Supervised Learning

What is Supervised Learning?

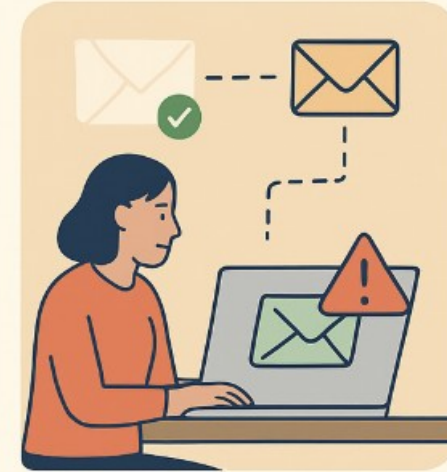
Supervised learning is a type of machine learning algorithm that involves learning from labeled data. In this approach, the algorithm is trained on a dataset that includes both input features (X) and corresponding output labels (Y). The goal is for the algorithm to learn the mapping between the input features and the correct output labels, enabling it to make accurate predictions.



Real-World Applications of Supervised Learning:

Supervised learning finds extensive applications in various domains. Here are a few examples:

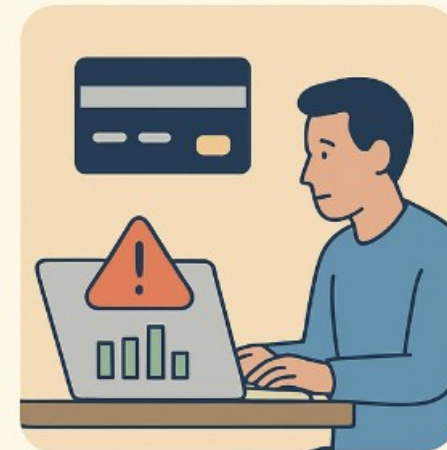
SUPERVISED MACHINE LEARNING



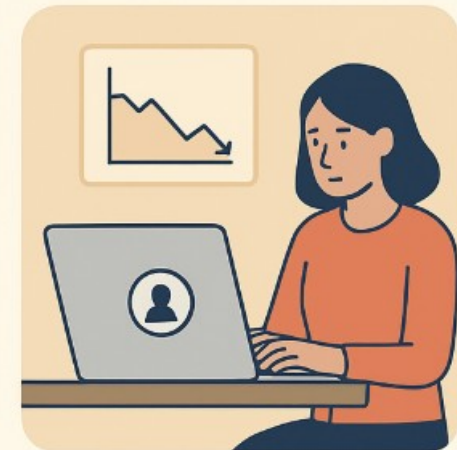
Email Spam Filter



Image Recognition



Fraud Detection



Customer Churn Prediction

Supervised learning can be further categorized into regression and classification tasks:

1. **Regression:** Regression algorithms aim to predict a continuous numerical value as the output. For instance, predicting housing prices based on features like size, location, and number of rooms.
2. **Classification:** Classification algorithms, on the other hand, predict discrete categories or classes. Examples include detecting whether an image contains a cat or a dog, or determining if a tumor is benign or malignant.

Housing price prediction(Regression task)

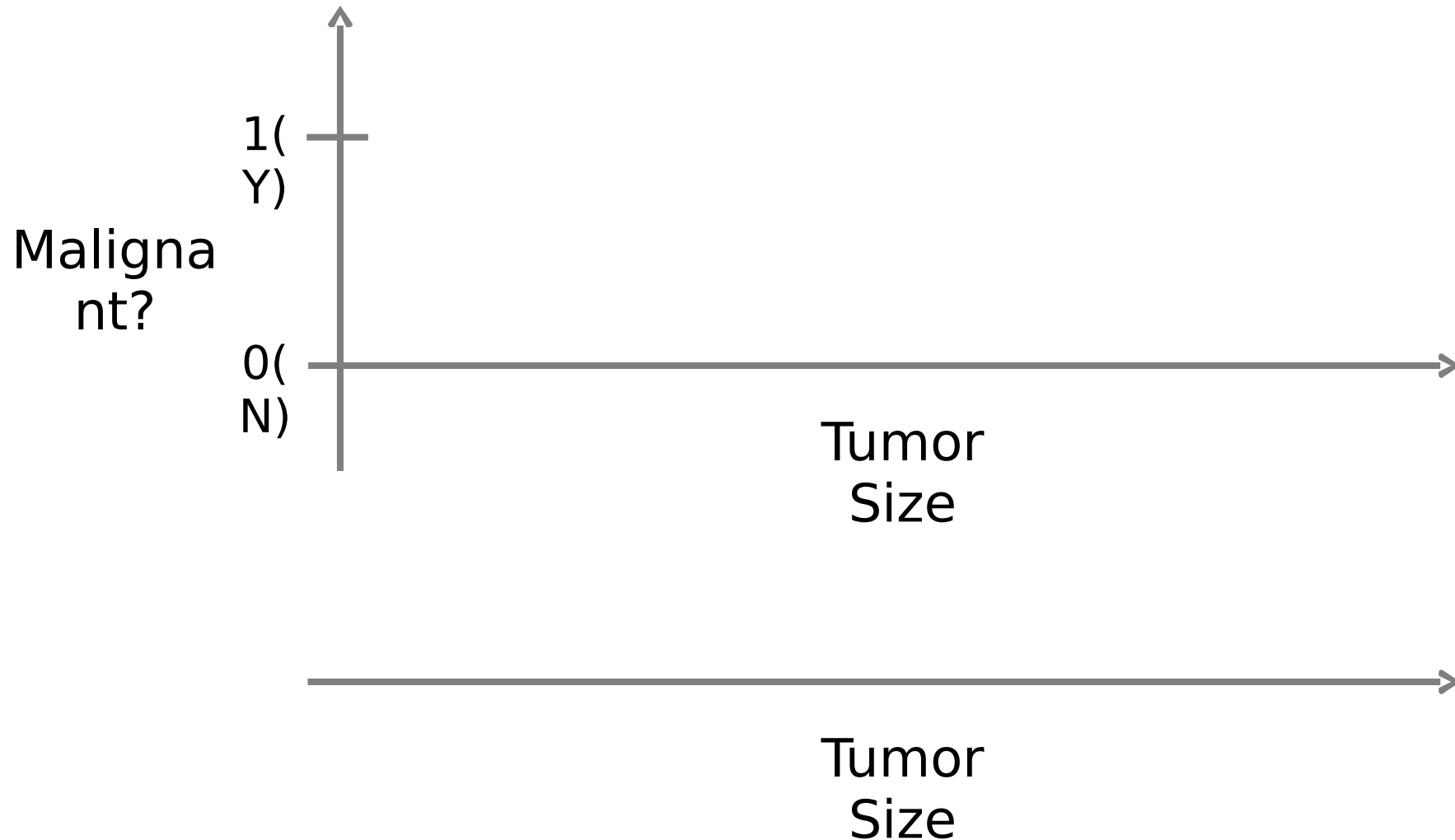
Price
(\$)
in
1000'
s

Size in
feet²

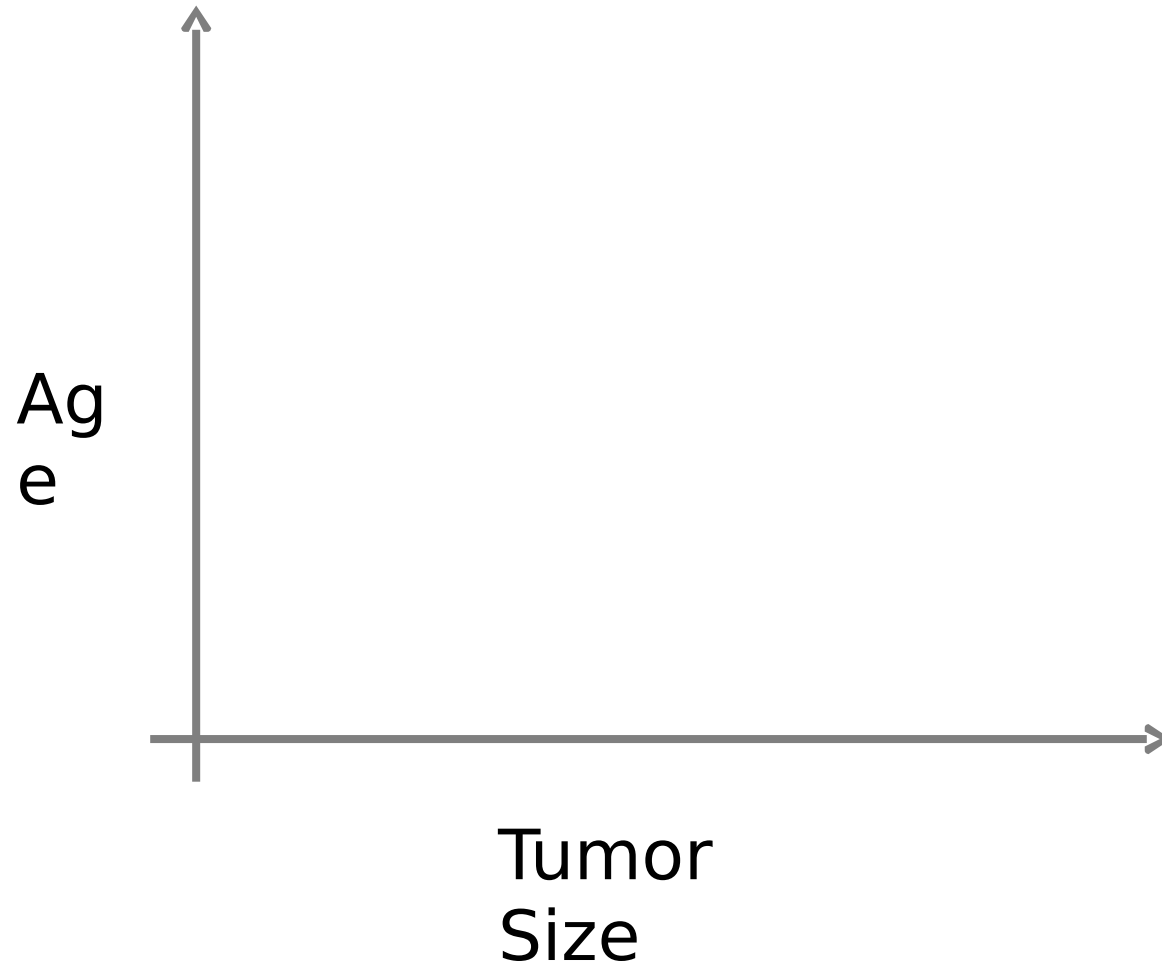
Supervised Learning
“right answers” given

Regression: Predict
continuous valued
output (price)

Breast cancer (malignant, benign)



Classification
n
Discrete
valued
output (0 or
1)



- Clump Thickness
- Uniformity of Cell Size
- Uniformity of Cell Shape

...

You're running a company, and you want to develop learning algorithms to address each of two problems.

Problem 1: You have a large inventory of identical items. You want to predict how many of these items will sell over the next 3 months.

Problem 2: You'd like software to examine individual customer accounts, and for each account decide if it has been hacked/compromised.

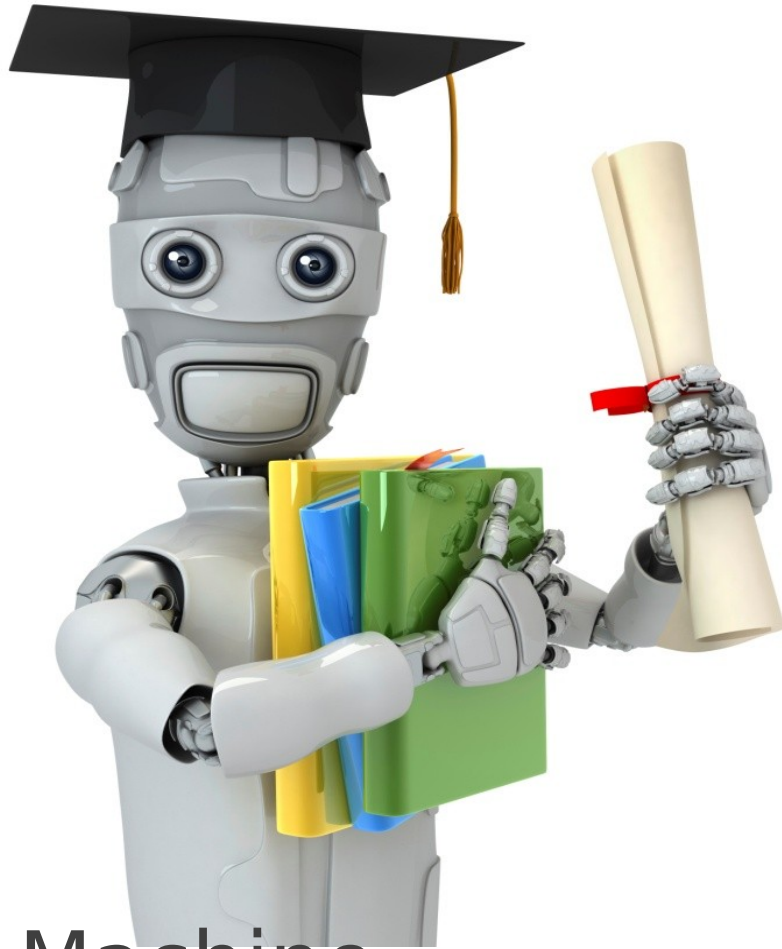
Should you treat these as classification or as regression problems?

Treat both as classification problems.

Treat problem 1 as a classification problem, problem 2 as a regression problem.

Treat problem 1 as a regression problem, problem 2 as a classification problem.

Treat both as regression problems.

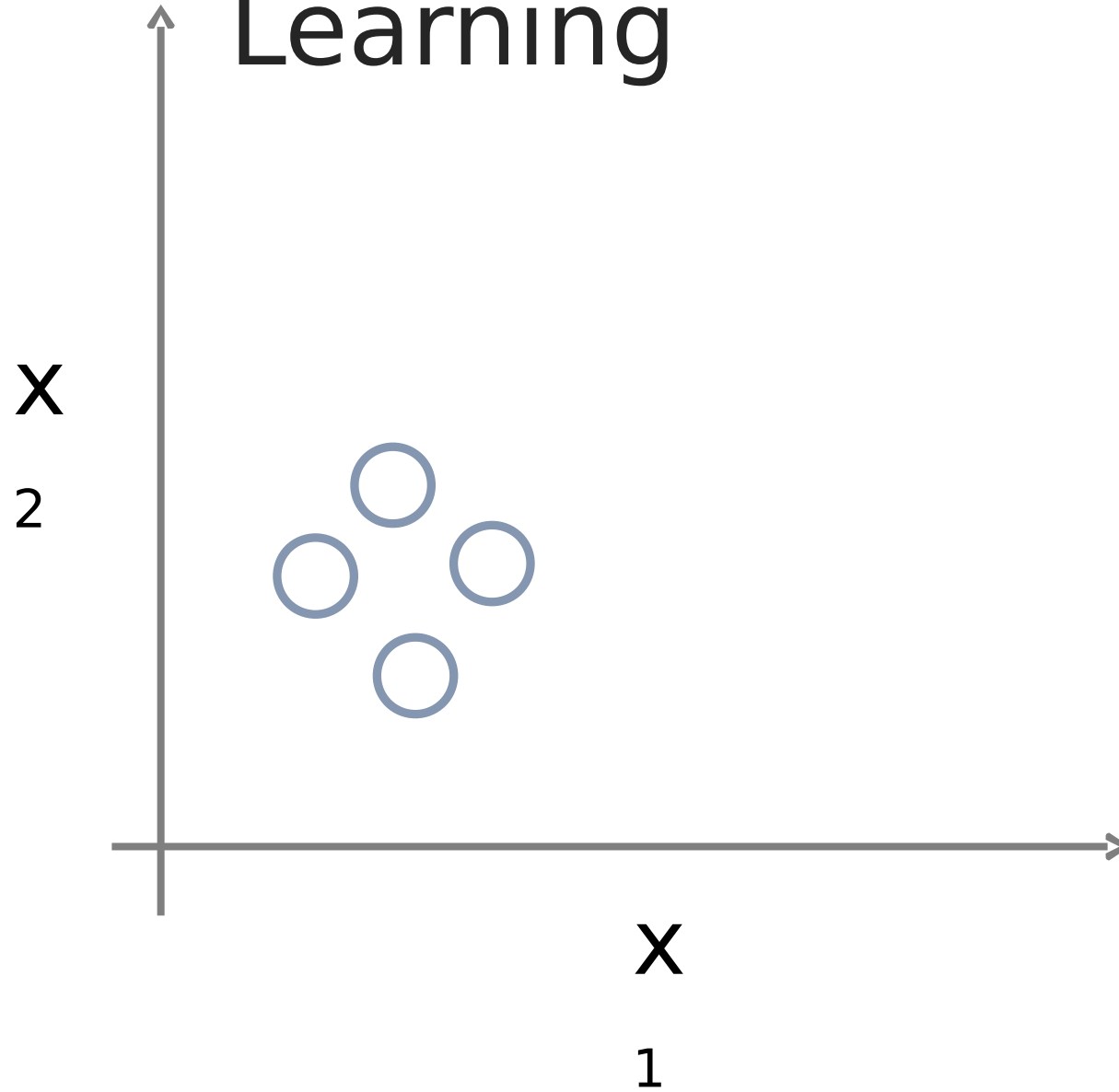


Machine
Learning

Introduction

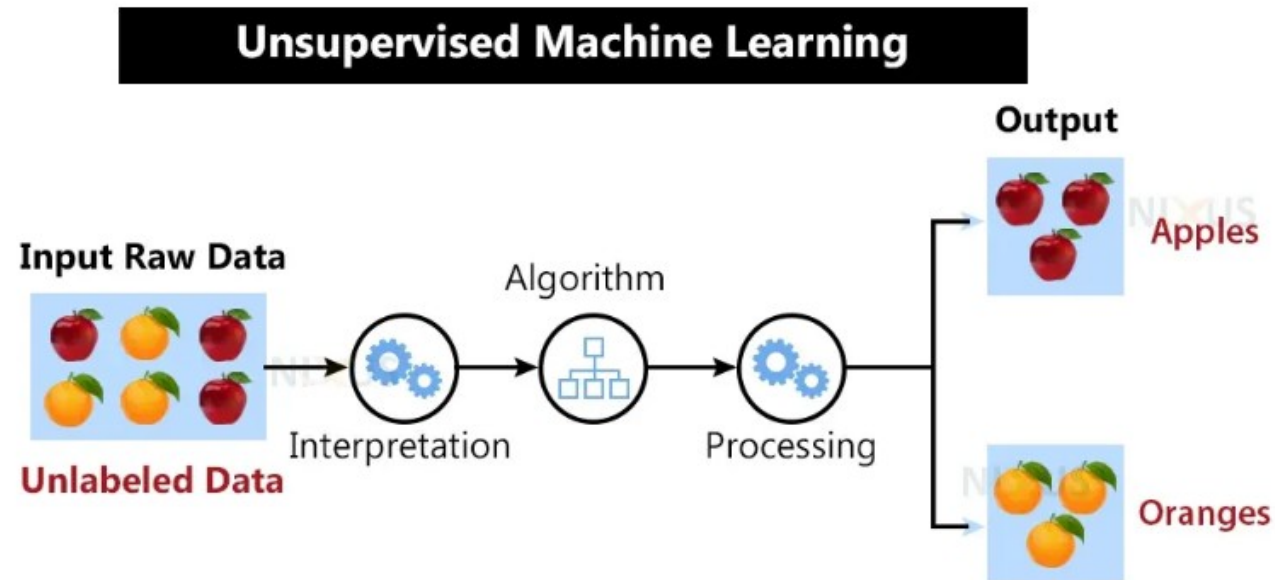
Unsupervised Learning

Unsupervised Learning



What is Unsupervised Machine Learning?

Unsupervised machine learning algorithms operate on unlabelled data, making it ideal for exploring and discovering patterns without prior knowledge or guidance. It helps uncover hidden structures, relationships, or clusters within the data, enabling deeper insights and better decision-making.



Of the following examples, which would you address using an unsupervised learning algorithm? (Check all that apply.)

Given email labeled as spam/not spam, learn a spam filter.

Given a set of news articles found on the web, group them into set of articles about the same story.

Given a database of customer data, automatically discover market segments and group customers into different market segments.

Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not.

Of the following examples, which would you address using an unsupervised learning algorithm? (Check all that apply.)

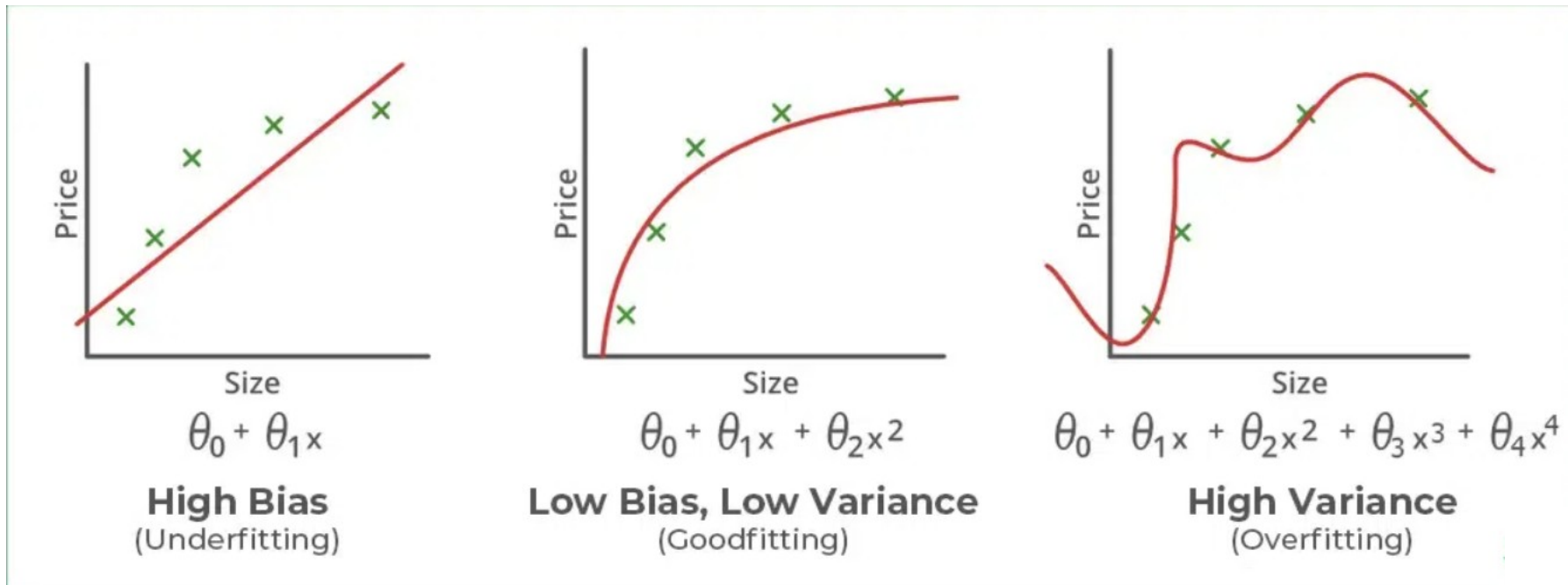
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Under-fitting Vs Over-fitting



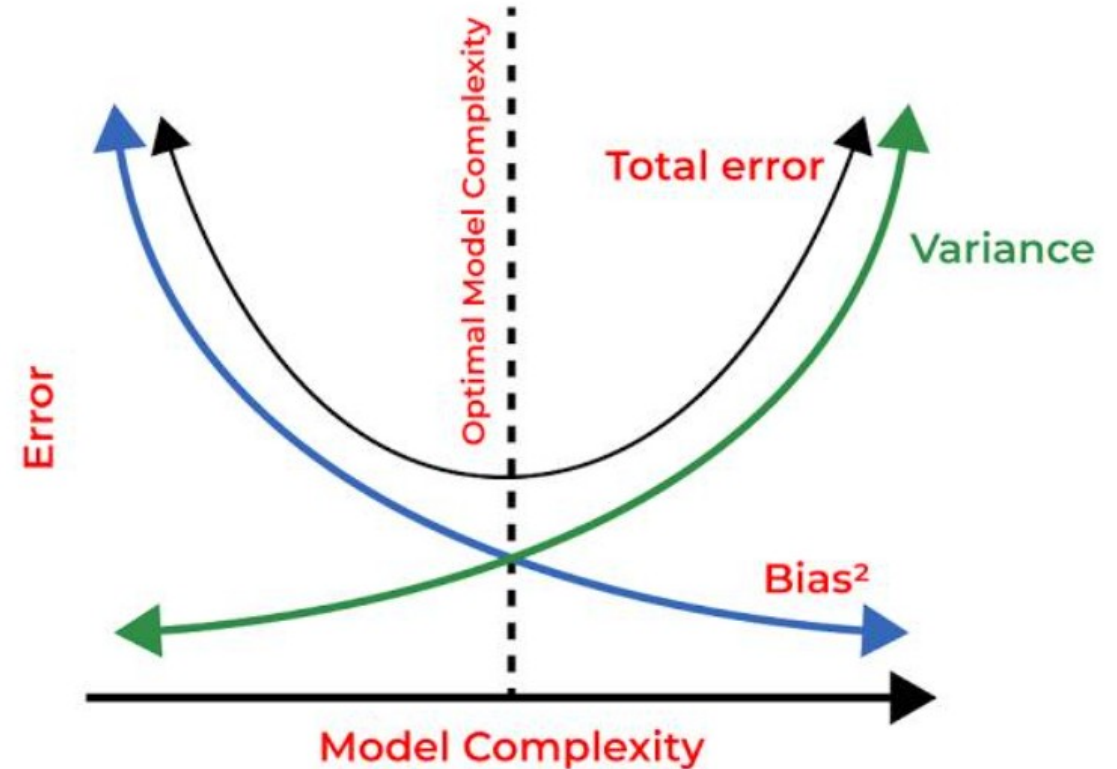
Under-fitting Vs Over-fitting

- **Underfitting** : Prediction line trying to fit a curved dataset but cannot capture the data's patterns, leading to poor performance on both training and test sets.
- **Overfitting**: Prediction curve passing through all training points, failing to generalize performing well on training data but poorly on test data.
- **Appropriate Fitting**: Curve that follows the data trend without overcomplicating to capture the true patterns in the data.

Bias-Variance trade-off

Bias is the difference between the average prediction of our model and the correct target value which model is trying to predict.

Low Bias is desired : High Bias means Under-Fitting of the model on training data.



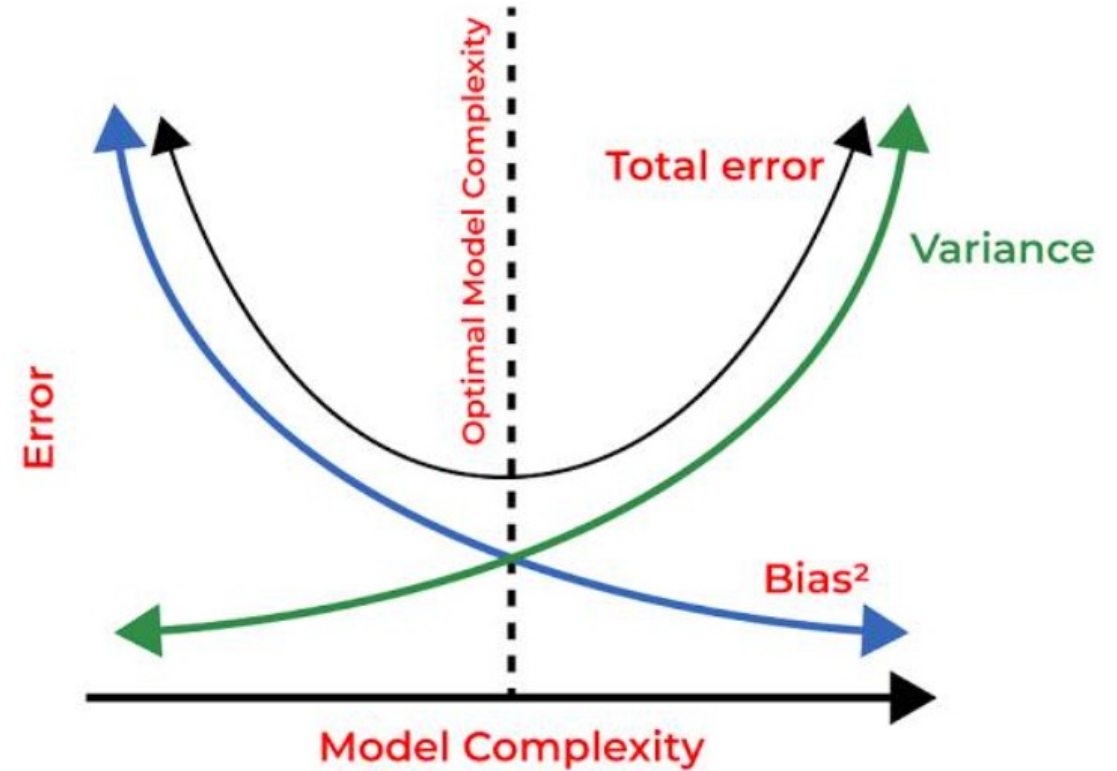
Bias-Variance trade-off

Bias is difference between the average prediction of our model and correct target value which model is trying to predict.

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Variance is the amount that the estimate of the target function will change if different training data was used.

Variance is error from sensitivity to small fluctuations in the training set. High variance can cause an algorithm to model the random noise in the training data, rather than the intended result



Bias-Variance trade-off

- When a model is too simple, like fitting a straight line to curved data, it has **high bias** and fails to capture the true relationship, leading to **underfitting**. For example, a linear model cannot represent a non-linear increase in house prices with size.
-
- However, if the model becomes too complex, like a fourth-degree polynomial that adjusts to every point, it develops **high variance**, overfits the training data, and struggles to generalize to new data. This is **overfitting**, where the model performs well on training but poorly on testing.
-
- An ideal model strikes a balance with **low bias and low variance**, capturing the overall pattern without overreacting to noise. For instance, a smooth second-degree polynomial fits the data well without being overly complex

THANK YOU

